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The Rowdy Robot | North America

Who Can Catch China's Humanoids?

China's 2nd annual World Humanoid Robot Games in Beijing showed the remarkable rate of change in humanoid development across locomotion, speed/power, fine motor skills and autonomy. Is China's physical AI lead over the US widening? When will the US have its first Robot Games?

- **Faster than Bolt.** A Chinese humanoid just ran the fastest 100 meters in history. Tiangong Ultra won the large-size 100-meter final in 8.64 seconds, beating both Usain Bolt's world record of 9.58 seconds and its own winning time of 21.5 seconds from last year's Games. The gain was achieved by reducing mass in the torso and waist, and upgrading the robot's motors and control stack. For context, humans needed roughly a century to cut 10% off the 100-meter record. Robots just cut 60% from the humanoid record in twelve months.
- **It's all in the hands.** The Dexterous Hand Challenge, a new competition in this year's Games, featured eight events: **1)** Power-tool assembly, **2)** Powder weighing, **3)** Bricklaying, **4)** Nail fastening, **5)** Lid opening, **6)** Box unpacking, **7)** Bean-picking with tweezers, and **8)** Cable connections. These events showcased recent humanoid developments in fine motor skills and autonomy, and they map directly onto value-add tasks in manufacturing and warehouse automation. They also exposed the gap between the leaders and the rest of the field, as some robots struggled to hammer a nail, while other robots demonstrated impressive dexterity. Moravec's paradox is clear, with robots demonstrating adult-level performance on the track, and a toddler's performance with their hands.
- **Is China's lead widening? Chinese vendors shipped more than 97% of global humanoids in H1 2026, a market itself up 272% Y/Y.** Chinese customers accounted for more than 85% of demand. Our China Industrials Team [recently lifted](#) its 2026 China shipment forecast to 50k units from 28k and sees 446k units by 2030e. Unitree, the first humanoid maker to list on the mainland, closed its Shanghai debut up ~460%, near a \$50bn valuation, though it has since given back ~45% to a market cap of Rmb190bn (~\$28bn).
- **PR win, cultural inertia.** Beijing just spent five days framing humanoids more as athletes than as job-takers, filling stadiums and drawing global press to a state-organized showcase of humanoid performance. We have argued that the industry's social license to deploy humanoids may matter as much as technical progress. State media now calls robotics, AI and innovative drugs the "new new three", successors to EVs, batteries and solar panels. Beijing has slotted humanoids into an export story that has worked across technology cycles, and the Games were the advertising campaign. A stadium of cheering fans tells you something about Chinese optimism on robotics

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that no shipment number can.

- **Who can catch them?** Forget the unit volumes for a moment. We view shipment share as the wrong scoreboard because the real gap is humanoid capability. The US still leads on frontier models, VLA development and simulation. China controls the physical inputs and the supply chain that enable scale. The question is whether US software leadership holds long enough, because every fleet China deploys is collecting the real-world data that could close the software gap too. Note that Unitree is allocating 48% of its IPO proceeds to model R&D, which tells you where the humanoid integrator thinks the binding constraint sits.
- **Consider the task list.** In March, we published a list of everyday tasks, from buttoning a dress shirt to repairing a Swiss watch, that would demonstrate extraordinary dexterity and cognitive ability in a humanoid. While nothing in Beijing cleared that bar, the Dexterous Hand Challenge moves in that direction, and the 100-meter record shows what happens to a benchmark once the industry decides to attack it.
- **See more insights** following the World Humanoid Robot Games from the Morgan Stanley China robotics team [HERE](#).

Exhibit 1: Recent Chinese Humanoid Policy

National Policies	Release Date	Target
MIT: Guiding Opinions On The Innovative Development Of Humanoid Robots	2023.10	By 2025, products reach internationally advanced levels and achieve mass production; cultivate 2–3 ecosystem companies with global influence
Standards System For Humanoid Robots And Embodied Intelligence	2026.2	Covers six pillars: basic commonality, brain/intelligent computing, limbs and components, robots and systems, applications, and safety and ethics
"15th Five-year Plan"	2026.3	Lists robots as a strategic emerging industry for the first time
2026 Humanoid Robots And Embodied Intelligence Real-world Training Special Initiative	2026.6	- By end-2026, humanoid robots complete application validation and routine deployment across a set of representative scenarios - Forming 100+ high-value applications, driving the formation of 10k-unit-scale deployment capability - At least 20 key scenario units per province, and 10 per central SOE
Local Government Policies	Release Date	2027 Target
Zhejiang Province Implementation Plan For Innovative Development Of The Humanoid Robot Industry (2024–2027)	2024.9	- Provincial annual complete-machine output reaches 20,000 units; - Core-industry scale of RMB20bn; related-industry scale RMB50bn
Shenzhen Embodied Intelligence Robot Technology Innovation And Industrial Development Action Plan (2025–2027)	2025.3	- Cultivate 10+ new companies valued over RMB10bn, 20+ companies with revenue over RMB1bn - achieve 50+ billion-yuan-scale application deployments, related-industry scale reaches over RMB100bn
Shanghai Embodied Intelligence Industry Development Implementation Plan	2025.7	- Expand the industry scale to RMB50bn 30% subsidy for model R&D, 20% for application projects, 50% for ecosystem projects
Beijing Economic-technological Development Zone Measures To Promote Innovative Development Of Embodied Intelligence Robots	2025.8	- Form 10,000-unit-scale mass-production capability for embodied intelligence robots - Build tens-of-millions-scale real-world embodied intelligence dataset
Hangzhou Embodied Intelligence Robot "Chain Strengthening" Action Plan (2026–2027)	2026.1	- Complete-machine enterprises' total industrial output exceeds Rmb20bn - Industry-chain total industrial output exceeds Rmb50bn

Source: Company Websites, Morgan Stanley Research

Exhibit 2: China Humanoid Value Chain



Source: Morgan Stanley Research

Exhibit 3: The Dexterous Hand Challenge Tasks Included 1) Power-tool assembly, 2) Powder weighing, 3) Bricklaying, 4) Nail fastening, 5) Lid opening, 6) Box unpacking, 7) Bean-picking with tweezers, and 8) Cable connections.



Source: Morgan Stanley Research

Exhibit 4: The China Humanoid Robot Games



Source: Global News

For Further Reading:

- [Humanoid Horizons: Moravec's Paradox](#) (1 Sep 2026)
- [Humanoid Horizons: Do Robots Have a PR Problem?](#) (28 Jul 2026)
- [The Rowdy Robot: Can it Change a Diaper?](#) (9 Mar 2026)
- [The Rowdy Robot: Special Theory of AI Relativity: \$i = wr^2\$](#) (23 Apr 2026)
- [The Rowdy Robot: Who's Gonna Fix the Clankers?](#) (28 Apr 2026)
- [Physical AI: The Morgan Stanley Robot Almanac](#) (7 Dec 2025)
- [Embodied AI: Who's Gonna Make the Bots? Adding Contract Manufacturers to the Humanoid 100](#) (26 Mar 2026)
- [The Rowdy Robot: Space 1493](#) (11 Mar 2026)

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	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
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Equal-weight/Hold	1577	43%	395	42%	25%	770	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	559	15%	93	10%	17%	216	12%
Total	3,694		939			1754	

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